

Predicting and monitoring local development from outer space

Evidence from the Bolivian municipalities

Carlos Mendez

Graduate School of International Development
Nagoya University



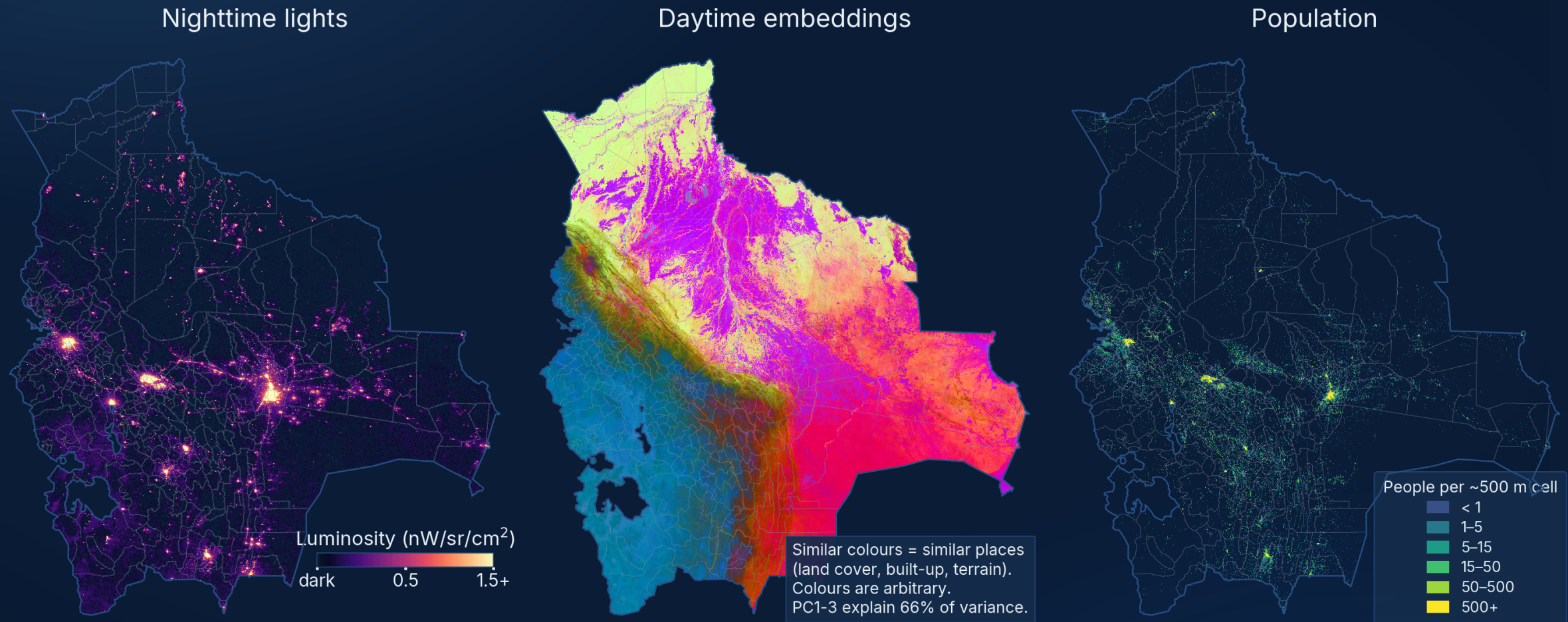
| Where development data is scarcest, it is needed most

THE PROBLEM

- A mayor planning a clinic has **no current poverty number** for their own territory.
- A census is a snapshot, not a series. Bolivia has **339 municipalities**, many too small or too remote to measure often.

The places that most need watching are the ones surveys reach last.

Lights, landscape, and people: three ways to see Bolivia



All three for 2017, on the same frame · VIIRS nighttime lights · AlphaEarth embeddings, first three principal components as red, green and blue · GHS-POP gridded population

| Development is visible from space — but which dimensions?

What is new

- Google's AlphaEarth turns a year of images into **64 numbers** for each patch of land.
- It learned on its own. Nobody showed it any social or economic data.
- The lights add **eight bands** of brightness for the same place.

What we ask

- We take **15 indices built from 62 indicators**, for all 339 municipalities.
- First: which goals can satellites *read*? Then: which can they *map*?
- A random forest, tuned by Bayesian optimization, learns the link.
- **Every number in this talk is out of sample.** No place helps predict itself.

Can satellites read development?

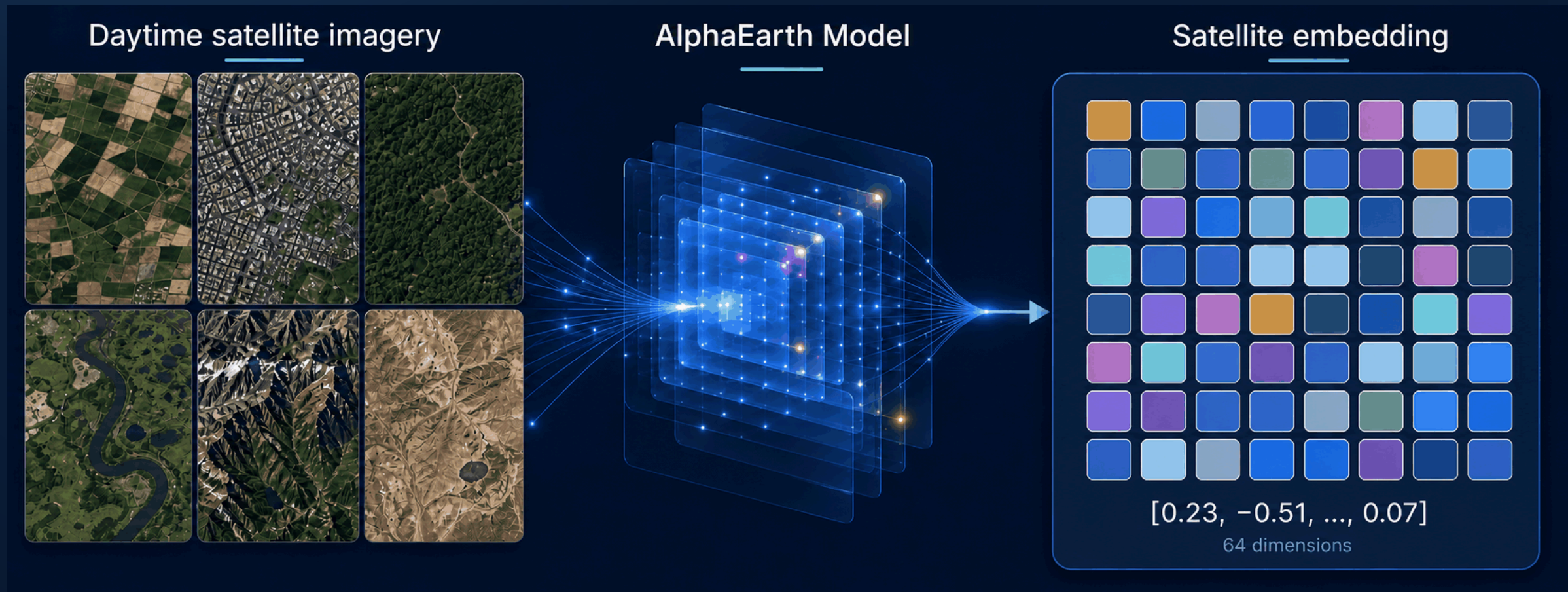
PART ONE · PREDICTION

Nighttime lights offer an opportunity to measure local development



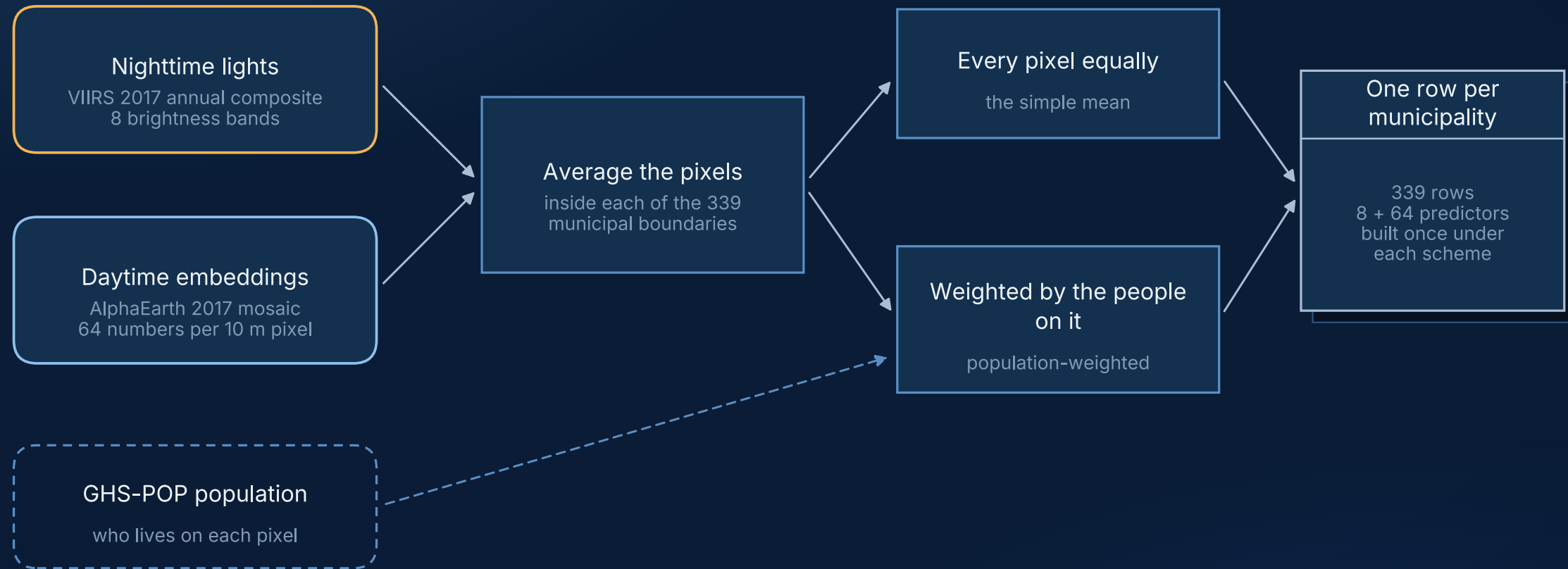
NASA Black Marble · Suomi NPP VIIRS · global context, not the study raster

Daytime embeddings: a new way to measure local development



Conceptual illustration · multiple daytime observations become one compact embedding

A satellite sees pixels; an index describes a municipality



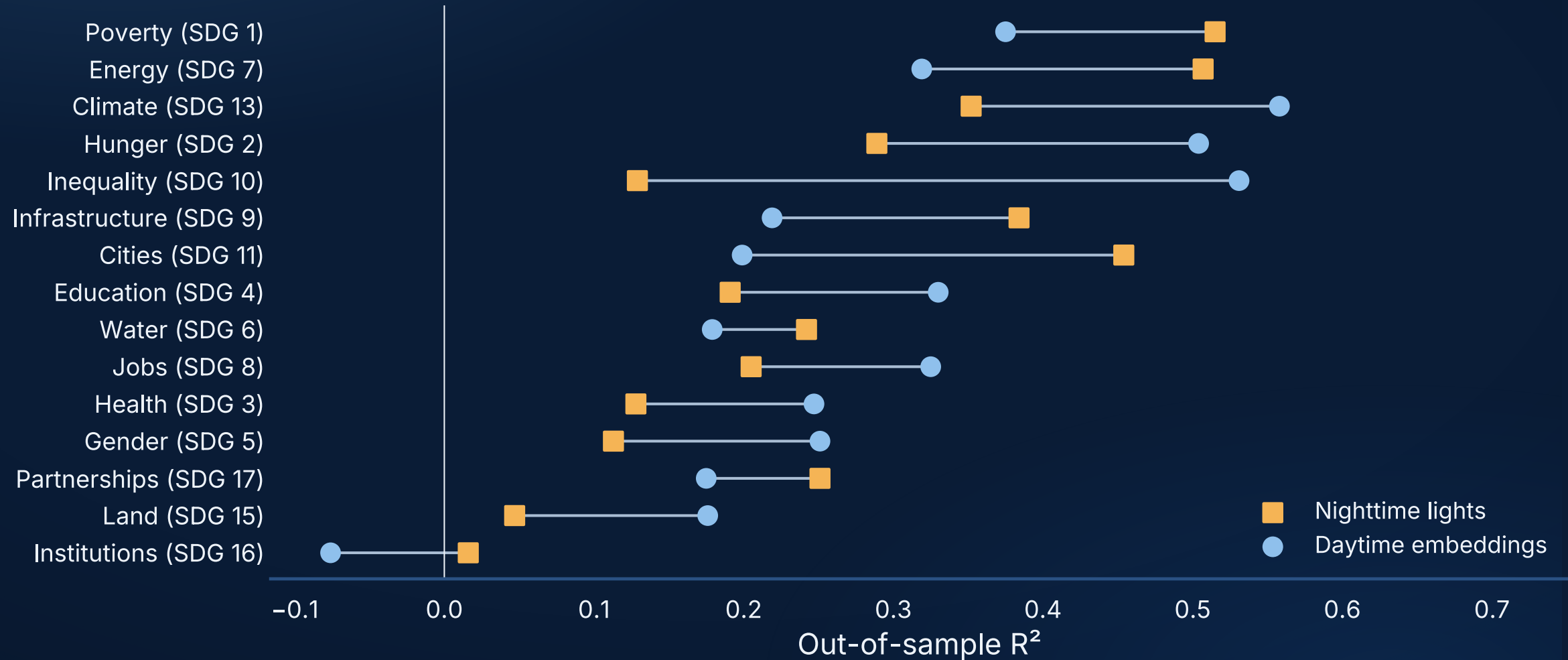
Rounded = a data product · sharp = a processing step · dashed = the population weights

| What a goal is made of decides if satellites see it

THE IDEA, BEFORE THE EVIDENCE

- Some goals are made of things you can **see**: housing, roads, land cover.
- Others are made of things you **cannot see**: health, gender, justice.
- So what a goal is made of should decide how well satellites read it.

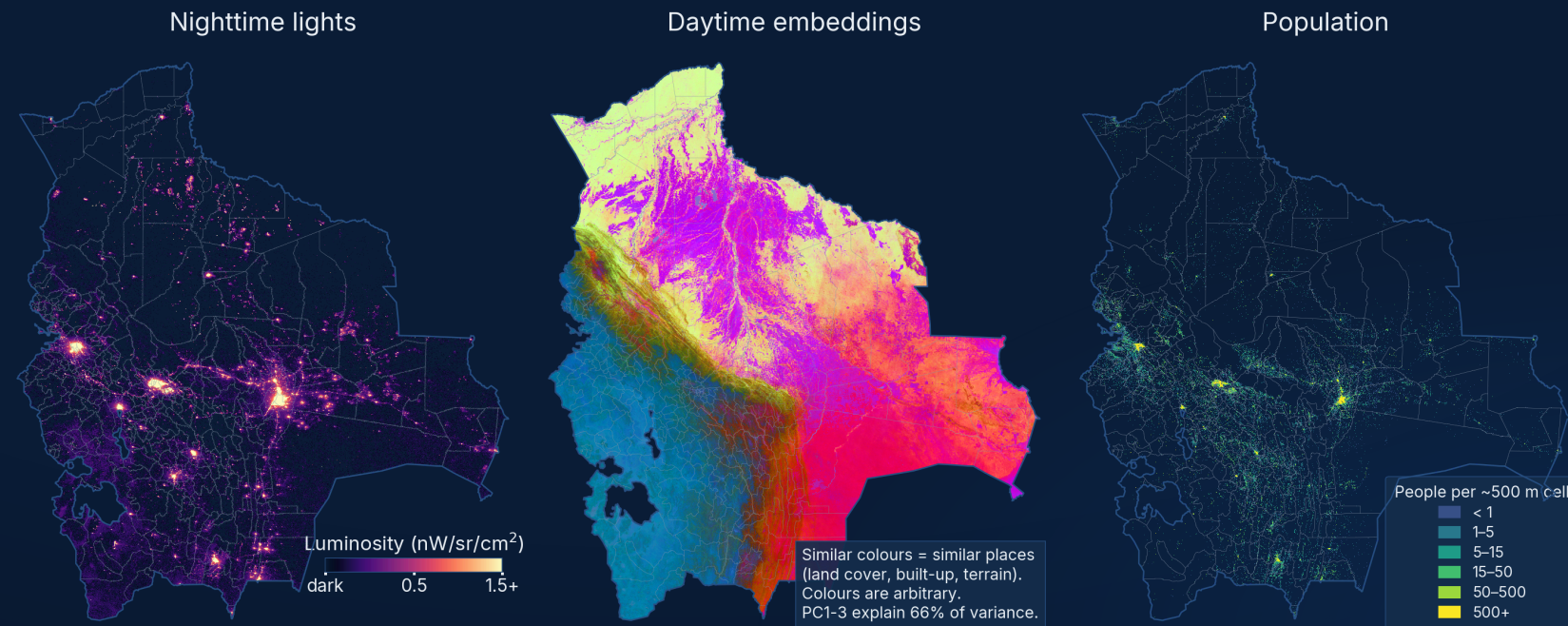
On the raw average, the lights win the goals we care about most



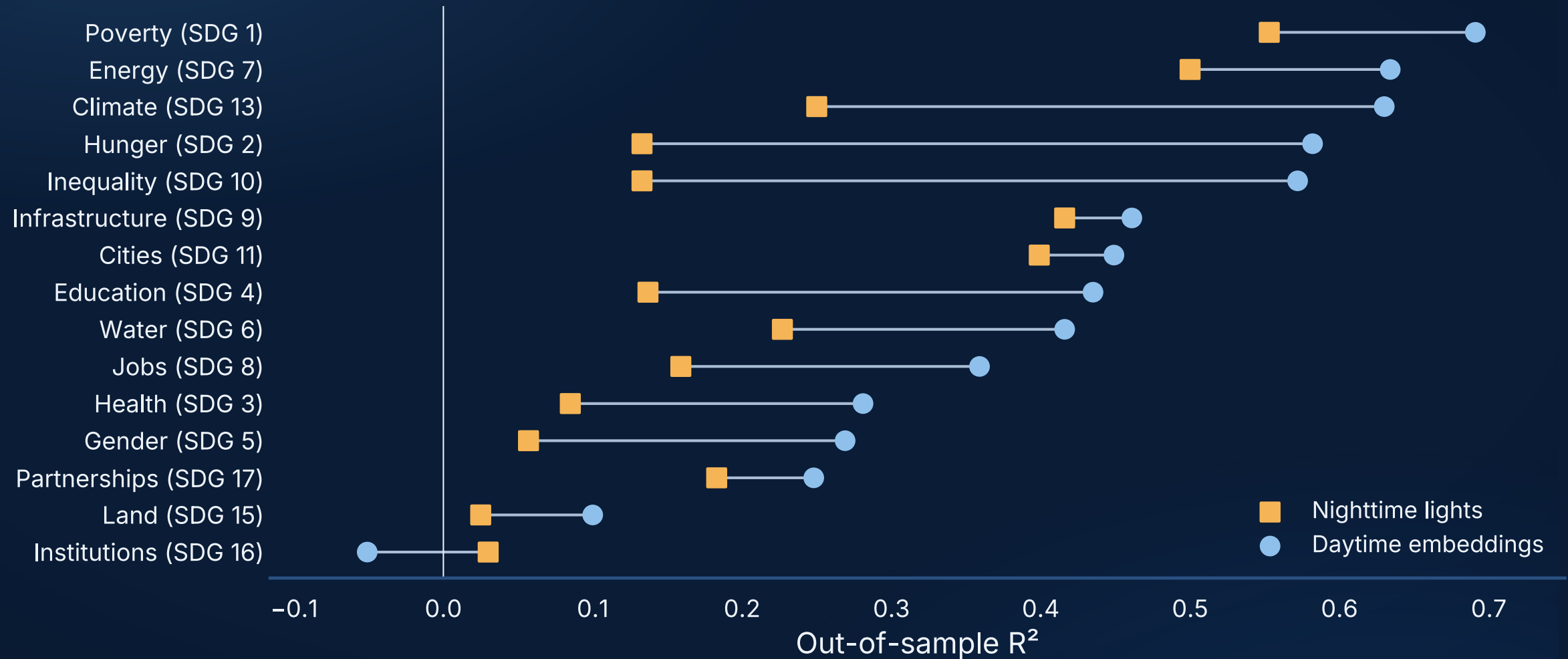
| Most of a municipality is empty — weight where people live

THE METHODOLOGICAL TURN

Weight each pixel by the people on it and the embeddings rise from **0.29 to 0.41**.
The lights do not gain — they slip from **0.25 to 0.22**.



Weight where people live, and the embeddings pull ahead

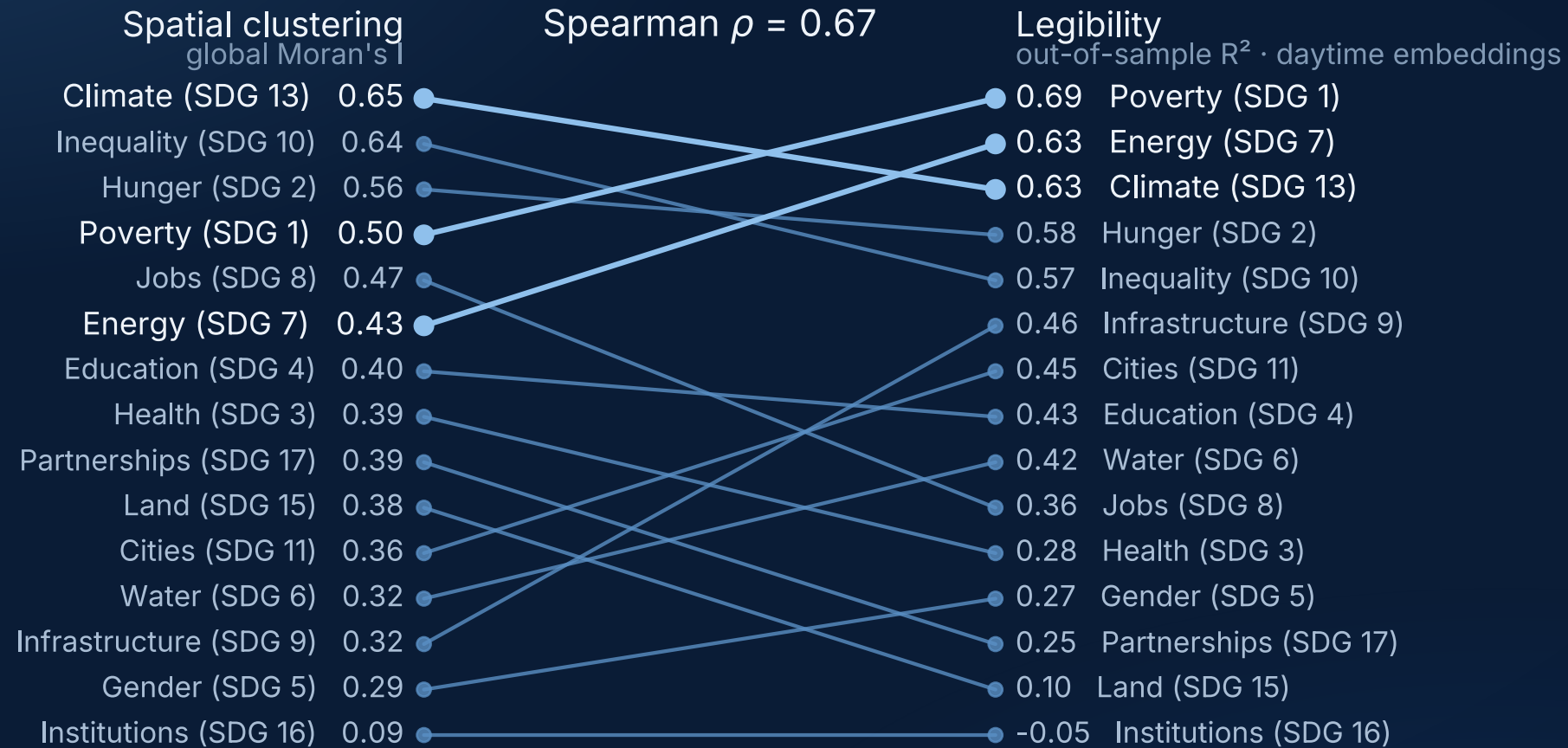


Weighted by population · out of sample · daytime embeddings now lead on 14 of 15 goals (8 → 14) — though the lead clears fold noise on only 5 of the 15

| The contribution is not one number — it is a map of legibility

- Written into the **physical fabric**, so read well: poverty 0.69, energy 0.63, climate 0.63, hunger 0.58.
- Assembled from **social and administrative records**, so barely read: institutions -0.05, gender 0.27, health 0.28.
- Reduced Inequalities, at 0.57, is the best-read of the socially defined goals.
- Life on Land, at 0.10, is the goal weighting *costs* us: its signal lives in the unsettled forest that weighting discards.

| Legibility has a spatial signature



The goals that cluster most in space are the goals the daytime embeddings read best · fifteen goals, ranked twice · ranked, not to scale

Which goals can we map — and how well?

PART TWO · MONITORING

| Together, the two views read poverty best of all

0.72

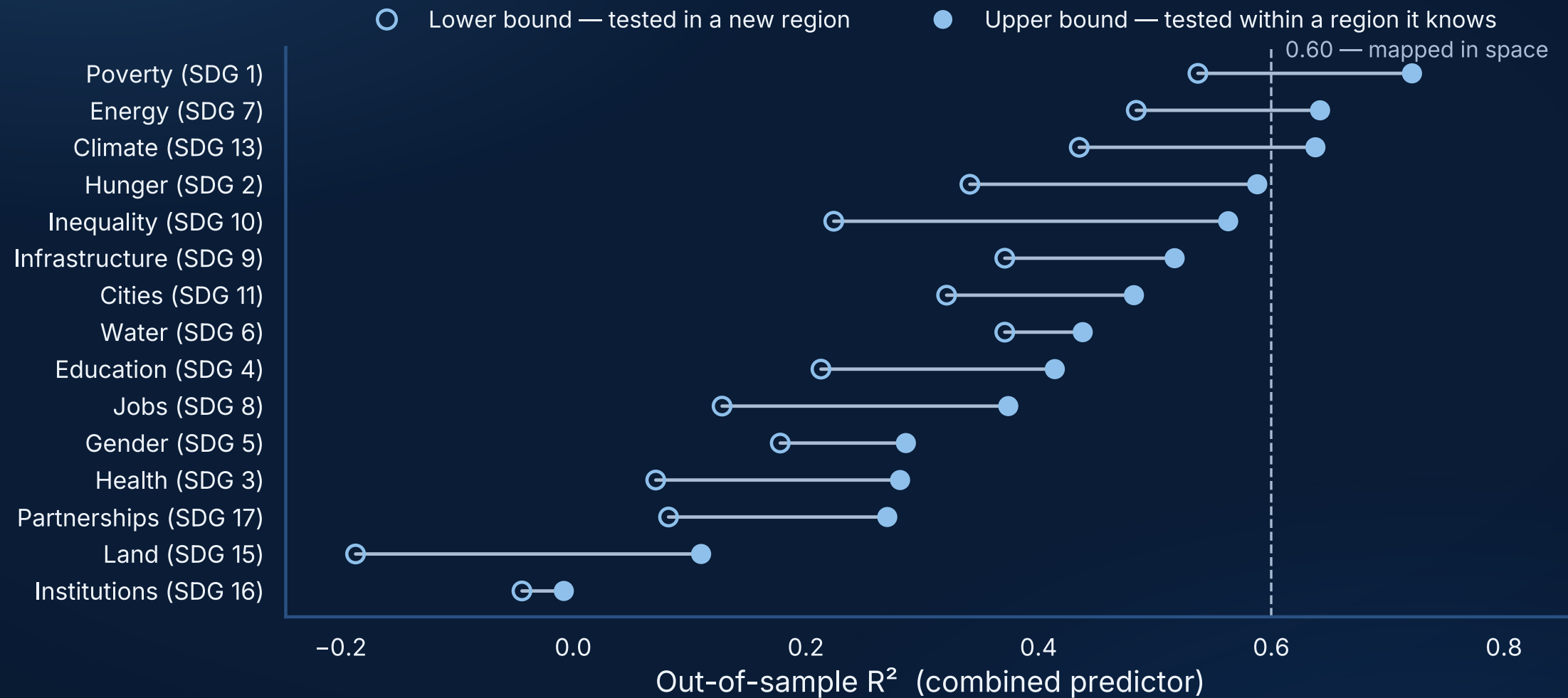
POVERTY · THE UPPER BOUND OF A 0.54–0.72
RANGE

We join the two views: **64 numbers** from daytime images and **8** from nighttime lights, both weighted by population. Together they beat either one alone on poverty.

Three goals pass **0.60**, our bar for drawing a map:

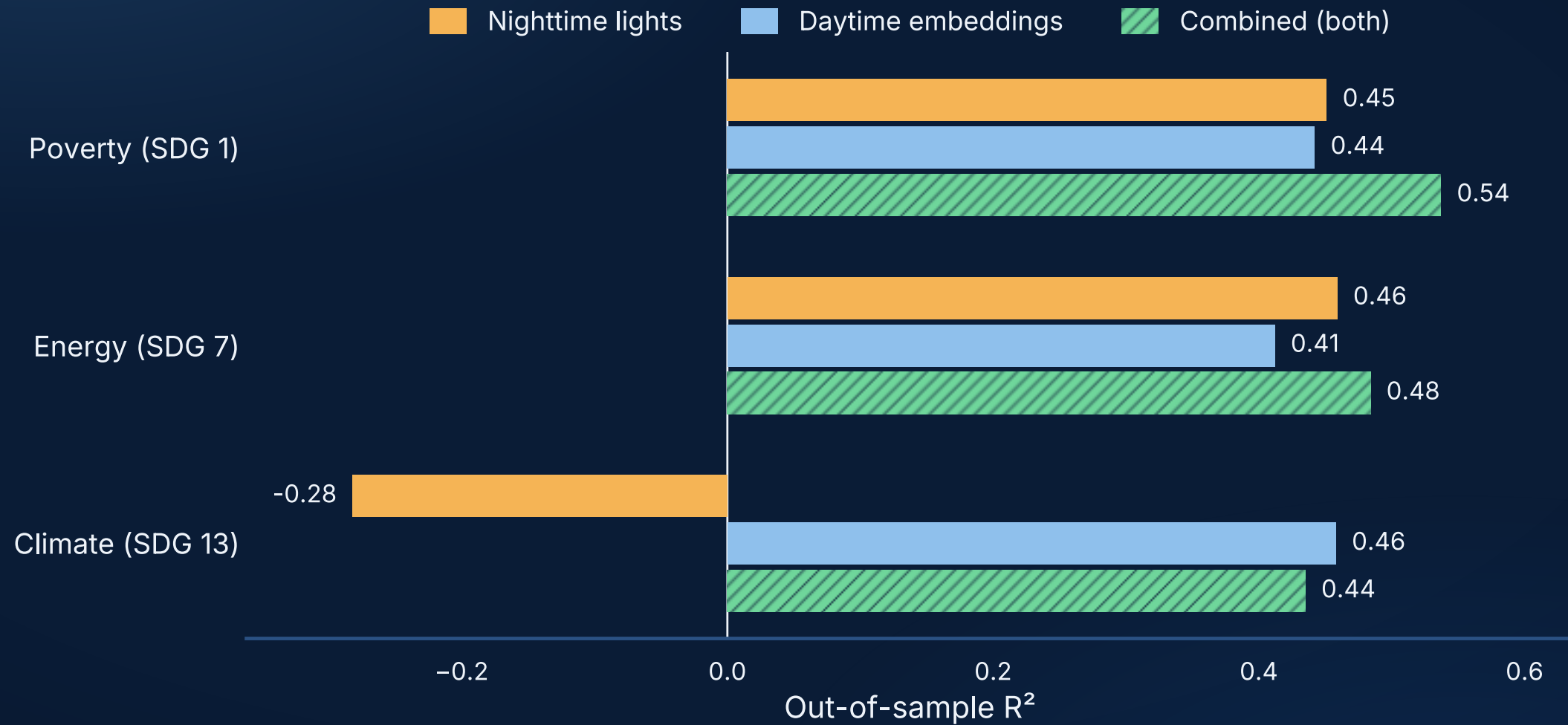
- Poverty (SDG 1) 0.72
- Energy (SDG 7) 0.64
- Climate (SDG 13) 0.64

| How you test decides what you get: 0.54 to 0.72



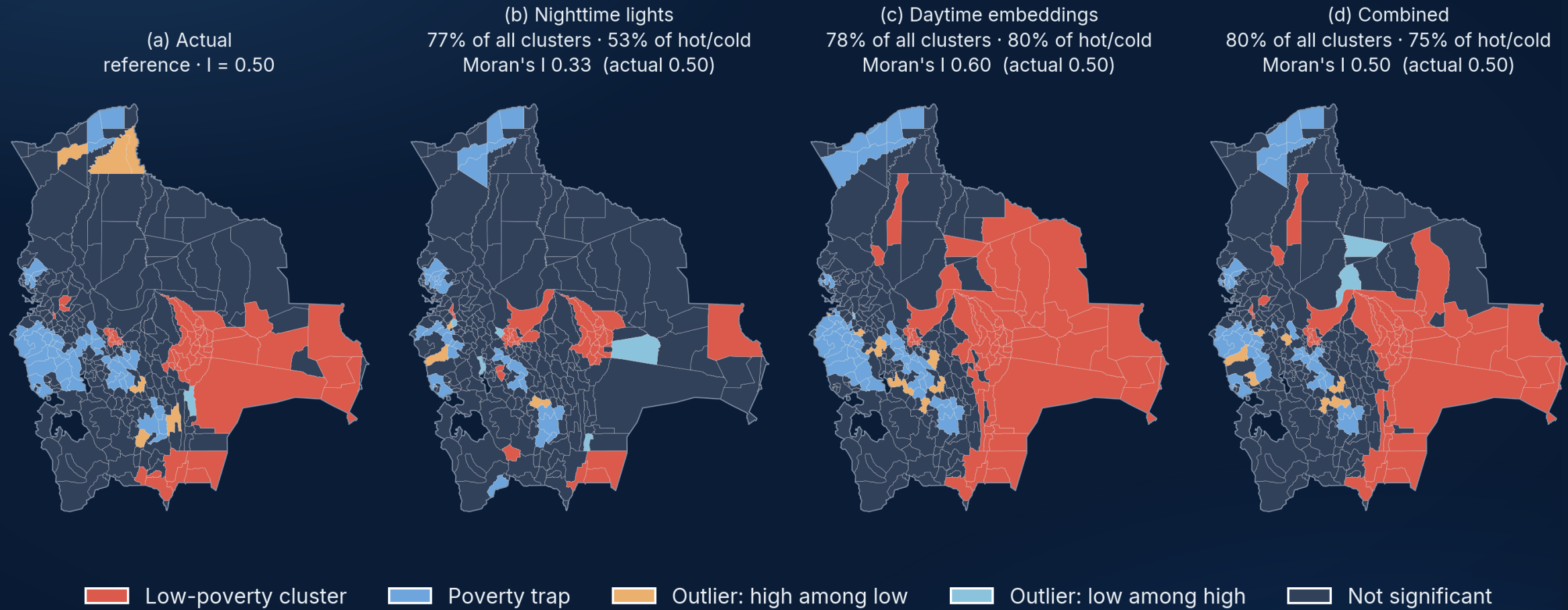
Each goal spans the two ways of testing · The three goals above the 0.60 line are the ones we map next

Under the hard test, only the fusion never collapses



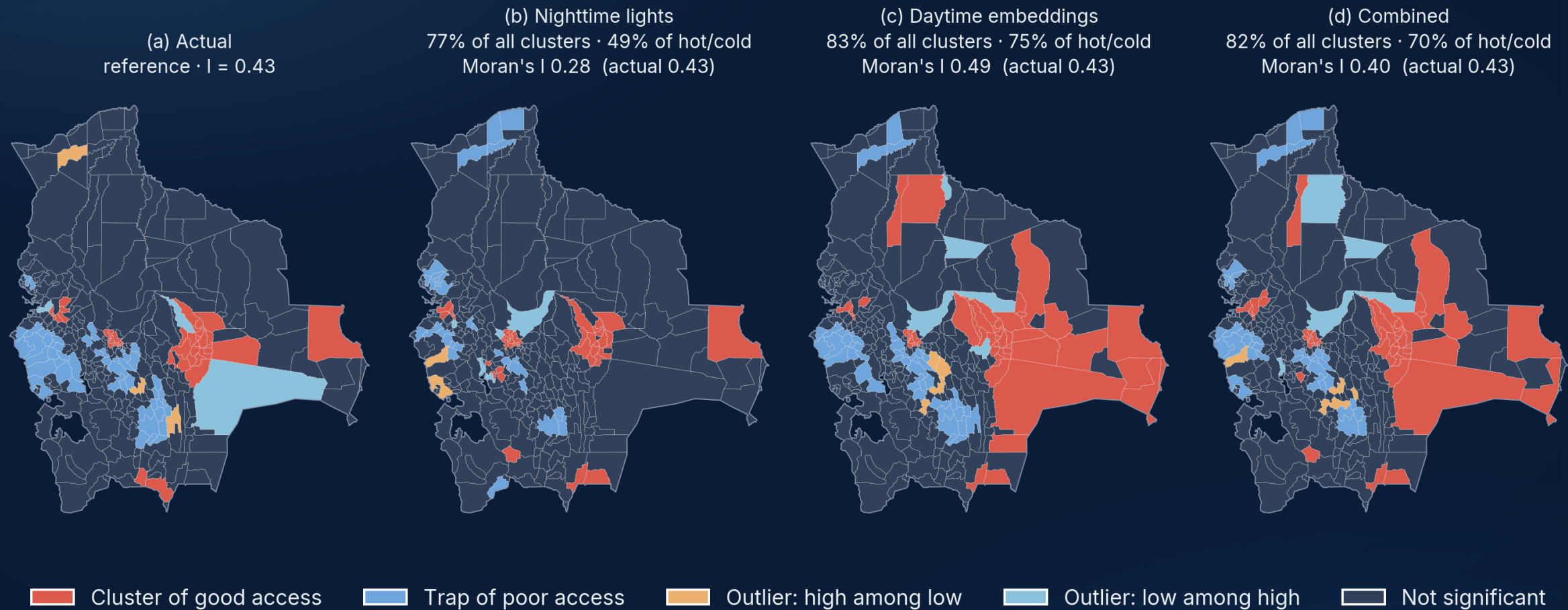
Tested in a department the model never saw — the bound that is not flattered by leakage

The combined view recovers the geography of poverty



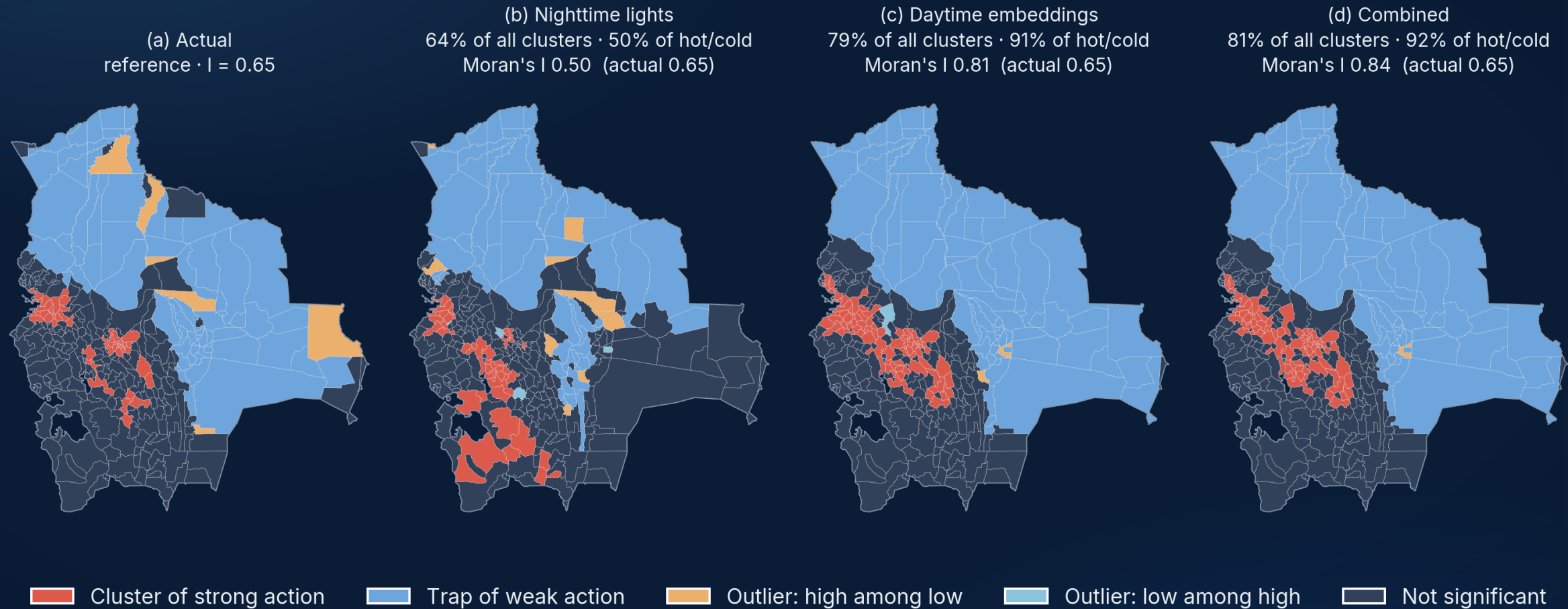
Red: clusters of low poverty · Blue: poverty traps · Each panel reports the share of the real hotspots and coldspots it finds

Clean energy: the lights read the level, not the pattern



Red: clusters of good access · Blue: traps of poor access · The lights reach $r = 0.73$ on the level, and half that on the pattern

Climate action: it finds the clusters — and overstates them



Red: clusters of strong action · Blue: traps of weak action · Recovers 92% of the real hotspots and coldspots — but makes the map look more clustered than it is

| The lead is real, but smaller than it looks

DEVIL'S ADVOCATE

- **"14 of 15 is a point estimate."** — True: it clears noise for only five goals.
- **"It depends on the test."** — In a new region the flip is $7 \rightarrow 8$, not $8 \rightarrow 14$.
- **"Nearby places leak in."** — Hence both bounds. Poverty does worse than the local mean in four departments of nine.
- **"Climate action is circular."** — Partly: its deforestation component is measured from space. Poverty is not.
- **"Combining always wins?"** — No: the embeddings alone find more hotspots on poverty and energy.

Satellites see the material geography of development

Let imagery track the goals it reads **between** survey rounds, and spend scarce survey effort on the goals it cannot read at all.

Together the two views read poverty at **0.54 to 0.72** out of sample and find its clusters and traps well enough to target scarce effort — but justice stays invisible, and health and gender barely register.

Validated on the 2017 cross-section. Carrying it forward between rounds is a projection, not a measurement.

Thank you

carlos-mendez.org